Pocket Option Market Data Harvester

Enterprise AI/ML Pipeline & Dashboard Operational Manual

Target Platform: Pocket Option (OTC & Global)	**Architecture:** FastAPI + WebSocket + Async Engine
Database Backends: Neon PostgreSQL (Cloud) + SQLite (Local)	**Alerting:** Automated Telegram Bot Service

1. System Purpose & Core Capabilities

This platform is engineered for **continuous 24/7 market recording** of high-frequency candlestick and tick movements across all currency pairs, with specialized support for **Over-The-Counter (OTC)** weekend markets. The data is rigorously processed for machine learning pipelines, algorithmic backtesting, and automated strategy training.

24/7 Zero-Loss Streaming	Automated Gap Recovery	Strict ML Integrity
Non-blocking asynchronous WebSocket collector with heartbeat ping keep-alive and exponential reconnect.	Self-heals after network stoppage by calculating missing time offset and backfilling historical bars in <500ms.	Strictly commits only finalized closed candles, eliminating lookahead bias and fixing wick anomalies automatically.

2. Web Dashboard Operational Guide

The web dashboard (accessible at http://localhost:8000) provides non-technical operators with full visual control over data harvesting, storage health, and streaming diagnostics.

Component	Function & How to Use
Master Controls	• **Start:** Boots the engine, connects to WebSocket, and begins recording. • **Pause / Resume:** Temporarily suspends recording without terminating socket. • **Stop:** Closes connection cleanly and flushes final bars to database.
Live Telemetry HUD	• **Status Badge:** Real-time operational state (RUNNING, PAUSED, RECONNECTING). • **Candles Total:** Cumulative closed candles stored in the database. • **Ingestion Rate:** Rolling candles processed per minute (speed gauge). • **WebSocket Latency:** Ping round-trip time in milliseconds (e.g., 25ms). • **Database Target:** Active storage backend (Neon PostgreSQL or SQLite).
Instrument & Period Selector	Allows dynamic multi-asset configuration (e.g., EURUSD_otc, USDARS_otc, GBPUSD_otc) and timeframe periods (60s for 1-minute, 300s for 5-minute). Changes take effect immediately without requiring a system restart.
Live Terminal / Log Stream	Streams raw WebSocket frames, heartbeat confirmations (42["ps"]), and backfill events directly to a scrolling dark-mode console window.
Recent Candlesticks Table	Displays the latest verified closed candles with microsecond timestamps, Open, High, Low, Close (OHLC), and tick Volume for instant audit.

3. Self-Healing & Gap Recovery Architecture

When an outage occurs (e.g. WiFi interruption, platform maintenance, or machine reboot), the engine executes an automatic 3-stage self-correction sequence:

Stage 1: Outage Detection — Upon reconnecting, the engine queries the active database: SELECT MAX(timestamp) FROM candles WHERE symbol = :sym AND timeframe = :tf.
Stage 2: Delta Calculation — It calculates gap_seconds = current_time - last_saved_time. If the gap exceeds 60s, it automatically requests missing bars using loadHistoryPeriod.
Stage 3: Idempotent Insertion — All missing bars (up to 1,440 per 24h block) are received in pre-calculated binary packages and inserted with ON CONFLICT DO NOTHING, guaranteeing zero duplicate records.

4. Telegram Notification System

The harvester connects directly to a dedicated Telegram bot to notify the trading team of all critical lifecycle events in real-time:

Event	Trigger	Alert Sample
■ **Harvester Started**	Engine booted via Web Dashboard or auto-restart.	■ Pocket Option Harvester STARTED • DB: PostgreSQL • Symbols: 4 • Timeframe: 60s
■■ **Reconnection Attempt**	Network disconnected; exponential backoff active.	■■ Harvester Disconnected Attempting reconnect #1 in 2.0s...
■ **Gap Recovered**	History backfilled after outage.	■ Reconnected & Recovered • EURUSD_otc: 120 bars patched
■ **Session Alert**	Platform auth rejected (token expired).	■ Session Expired (41 Unauthorized) Please refresh token in Dashboard.

5. Value-Add Roadmap: Suggested Future Features

To maximize the commercial and operational impact of the harvester for the client's algorithmic trading, the following non-intrusive enhancements are recommended for immediate presentation:

1. Live Lightweight Candlestick Chart in Dashboard

Embed a TradingView-style interactive HTML5 candlestick canvas directly inside the web dashboard. Operators can visually follow candles forming in real-time, inspect OHLC levels, and verify data parity against the Pocket Option trading terminal with zero performance overhead.

2. One-Click AI/ML Dataset Export (CSV / Parquet / JSON)

Add an export drawer allowing users to filter by date range, select specific assets, and download pre-cleaned data files formatted specifically for Python Pandas, PyTorch, TensorFlow, and Backtrader backtesting scripts.

3. Data Completeness & Quality Scorecard (99.99% Gauge)

Display a visual health meter showing data completeness: total expected minutes vs. recorded bars, average ping latency, and zero-gap verification. This provides the client with proof of data integrity.

4. Automated 24-Hour Midnight Telegram Digest

A daily automated report card delivered to Telegram at 00:00 UTC summarizing total volume recorded, highest volatility pair, network uptime percentage, and cloud storage growth.

5. Multi-Timeframe Virtual Resampler (Virtual 5m / 15m / 1h)

Using the granular 1-minute closed bars, an in-memory aggregation module can generate higher timeframe candles on demand without opening additional WebSocket streams to Pocket Option.